

The new specific engineering intervention is the integration of a real-time edge computing platform that processes data from various sources and feeds it into the adaptive ensemble model, which then adjusts the EV's powertrain configuration in real-time.

Executive Technology Brief

Domain: ml_data

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'underperformance' concerning electric vehicle. User-submitted engineering problem (session input): "Create an electrical vehicle faster than Tesla Model Y and more efficient than BYD.". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

Mechanism: The new causal mechanism is a real-time adaptive ensemble machine-learning model that incorporates diverse data sources, including sensor data, weather forecasts, traffic information, and reinforcement learning algorithms for adaptive battery management.

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: The new specific engineering intervention is the integration of a real-time edge computing platform that processes data from various sources and feeds it into the adaptive ensemble model, which then adjusts the EV's powertrain configuration in real-time.

Governing Model: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: APPLICABLE; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: $P(x|y)$.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: APPLICABLE; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution

shift). Engaged variables: n , $f(x_i)$.

ML-001: Expected utility of a decision rule. Expression: $EU = \sum_y P(y | x) * U(a(x), y)$. Applicability to this invention: APPLICABLE; holds under: utility table fully specified; probabilities calibrated.

Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: EU.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.